MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS), ROCKET CATAPULT MARK 18 MOD 0 (LOADED ASSEMBLY)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Mark 18 Mod 0 Rocket Catapult described herein shall consist of this document and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Catapult Mk 18 Mod 0 (Loaded Assembly).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents are in addition to those listed in the cover specification, MIL-A-85097, and form a part of that specification to the extent specified herein.

SPECIFICATIONS

Naval Air Systems Command (Code Ident 30003)

AS 3531	Cartridge Propellant, N-45, Assemblies for Rocket Catapults Mark 12 Mod 1 and Mark 18 Mod 0
AS 3532	Propellant, N-46, Loaded Motor Tube Assembly For Rocket Catapults Mark12 Mod 1 and Mk 18 Mod 0

STANDARDS

Military

MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-2100	Propellant, Solid, Characterization of (Except Gun Propellant)

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL707AS100	Rocket Catapult Assembly Mark 18 Mod 0
707AS100	Rocket Catapult Assembly Mark 18 Mod 0
707AS129	Container Assembly For Rocket Catapults

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet unless in direct conflict with the general specification shall be considered additional, not substitute. The rocket catapult shall conform to DL 707AS100.

3.2 Preproduction/production.

3.2.1 Propellant. The propellant shall conform to the requirements of AS 3531 or AS 3532, whichever is applicable, when tested in accordance with MIL-A-85097.

3.2.2 Ballistic performance. The ballistic performance shall conform to table I. (See 6.2 for definitions.)

TABLE I. Ballistic performance requirements.

Parameter	306 lb −40° ± 5°F		306 lb 77° ± 5°F		306 lb 165° ± 5°F		374 lb −40° ± 5°F		374 lb 77° ± 5°F		374 lb 165° ± 5°F	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Catapult igniton delay (ms)	315	411	229	380	222	339	315	411	229	380	222	339
Maximum acceleration (g)	-	14.0	-	18.0	-	20.0	-	14.0	-	16.5	-	20.0
Maximum onset rate (g/s)	-	218	-	230	-	350	-	245	-	220	-	350
Maximum velocity (fps)	37	-	39	-	45	-	37	-	38	-	44	-
Rocket ignition delay (ms)	-	18	-	18	-	16	-	18	-	18	-	16
Maximum rocket resultant thrust (lb)	2250	4000	2220	5210	2850	6000	2250	4000	2220	5210	2850	6000
Rocket action time (ms)	375	540	268	565	268	397	375	540	268	565	268	397
Rocket impulse (lb-s)	757	-	770	-	811	-	757	-	770	-	811	-

3.2.2.1 Velocity. The velocity of the ejected weight at end of the catapult stroke, shall be measured over the interval from 28 to 40 inches of travel.

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3.2.2.2 **Propellant tests.** Propellant samples shall meet the requirements of AS 3531 or AS 3532, whichever is applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction.

4.1.1 **Sample size.** Twenty (20) rocket catapults shall be submitted for preproduction inspection.

4.1.2 **Inspection routine.** The sample shall be ballistically fired in accordance with the schedule specified in table II. Propellant testing shall be in accordance with Method 802.1 (Heat of Explosion) of MIL-STD-286, the uniaxial mechanical property test of MIL-STD-2100 and the ballistic propellant tests in accordance with AS 3531 or AS 3532, as applicable.

TABLE II. Ballistic firing test schedule.

Temp(°F)	Eject. wt.(lb)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
-40	306	X	X									X				X				X	
-40	374			X	X									X			X				
77	306					X															
77	374						X														
165	306							X	X				X					X			X
165	374									X	X				X				X		

4.1.3 **Temperature conditioning.** The catapult shall be temperature conditioned for 6 hours minimum as follows:

a. Cold $-40° \pm 5°F$
b. Hot $165° \pm 5°F$
c. Ambient $77° \pm 5°F$

4.2 **Production.** Production inspection shall consist of ballistically firing samples, selected as specified in Table III. Temperature conditioning shall be as specified in 4.1.3.

TABLE III. Ballistic firing sampling plan.

Lot size	Sample size	Test sample distribution at each condition			
		Temperature (°F)			
		-40	-40	165	165
		Ejected weight (lb)			
		306	374	306	374
Up to 40	6	2	1	2	1
41 to 65	8	2	2	2	2
66 to110	10	3	2	3	2
111 to 180	14	4	3	4	3
181 to 300	20	5	5	5	5

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5. PACKAGING

5.1 Packing. Rocket catapults shall be packed as specified on Drawing 707AS129.

6. NOTES

6.1 Intended use. The Mk 18 Mod 0 Rocket Catapult is intended to provide the propulsion force needed to eject personnel from the T-2 Aircraft.

6.2 Definitions.

6.2.1 Rocket motor resultant thrust. The rocket motor resultant thrust is obtained by vectorially adding the two maximum components of rocket thrust measured 20 msec or later after rocket motor ignition.

6.2.2 Rocket motor ignition time. Rocket motor ignition time is the elapsed time from strip-off(the point on the catapult thrust table where the table first begins to drop sharply) to 10 percent of maximum thrust, as measured on the rising portion of the normal component rocket motor thrust.

6.2.3 Double tang lock expulsion. Double tang lock expulsion is expulsion of the double tang lock from the catapult tube. This is an acceptable condition which can be randomly expected.

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